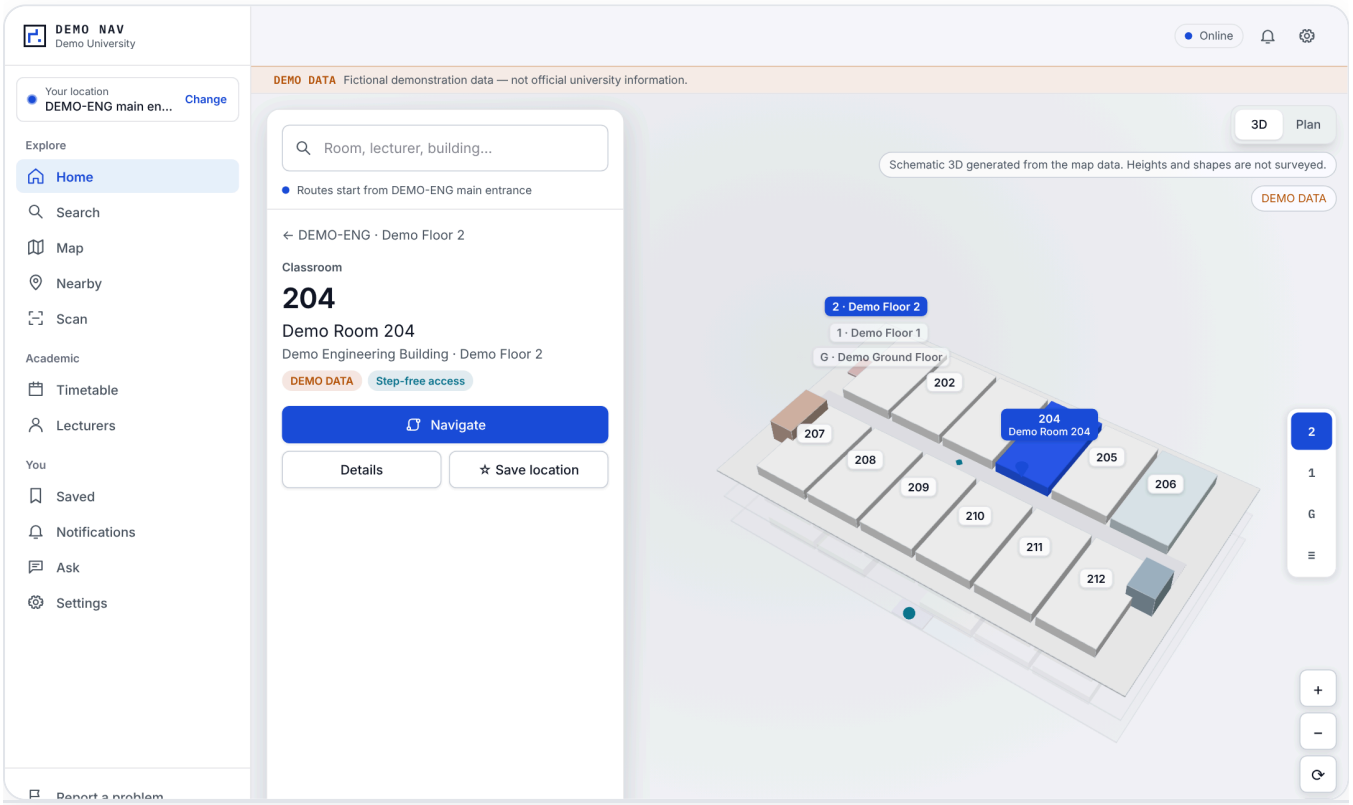
SUL NAV
Campus navigation for Salahaddin University-Erbil

A proposal to give every student, lecturer and visitor a simple way to find any building, room or office — on their phone, in Kurdish, Arabic or English, and even without internet.



Prepared for
Presidency, Salahaddin University-Erbil

Prepared by
NULL-TECH (project owner)

Status
Working prototype · pilot proposal · November 2026

1 · SUMMARY

What we are asking for, in one page

New students, visitors and even staff lose time every day looking for lecture halls, offices and services. **SUL NAV** is a campus navigation app that shows where things are and how to walk there, step by step, indoors and across floors.

Find

Search any room, lecturer office, department or building in Kurdish, Arabic or English.

Walk there

Turn-by-turn directions, including stairs, elevators and step-free routes.

Stay informed

Timetable, room changes, closures and announcements in one place.

Where the project is today

- **The software is built and tested:** student web app (works on any phone, installable, offline), staff administration dashboard, secure database design, 2D/3D maps, real-world map, QR checkpoints, timetable and three languages. Automated tests cover the routes, security rules and the main user journeys.
- **It currently runs on fictional demonstration data.** No Salahaddin University buildings, rooms or timetables have been entered, because that must be done with the university's official information and permission.
- **It has not yet been used by real students or tested at full university scale.** This proposal includes the plan to do that safely.

What we need from the university

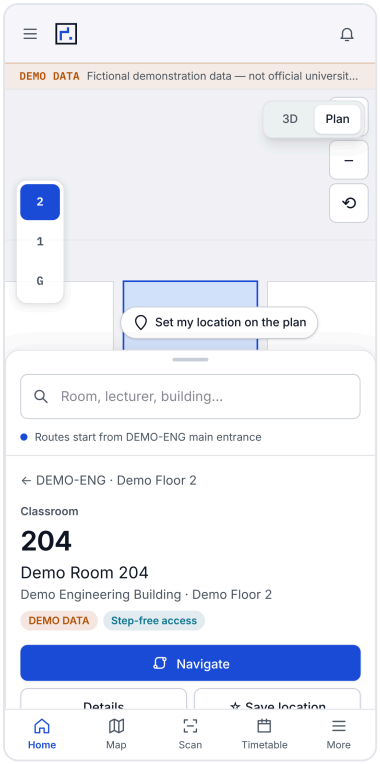
1. **Official approval** and one **coordinator** from the university who can open doors to the right offices.
2. **Campus information:** official building and room lists, floor plans, and confirmation of building locations.
3. **Permission** to put small QR signs at entrances and stairwells, and a short field survey to check every route.
4. **IT support:** an official web address (for example *nav.su.edu.krd*) and agreement on hosting and data policy.

Proposed next step

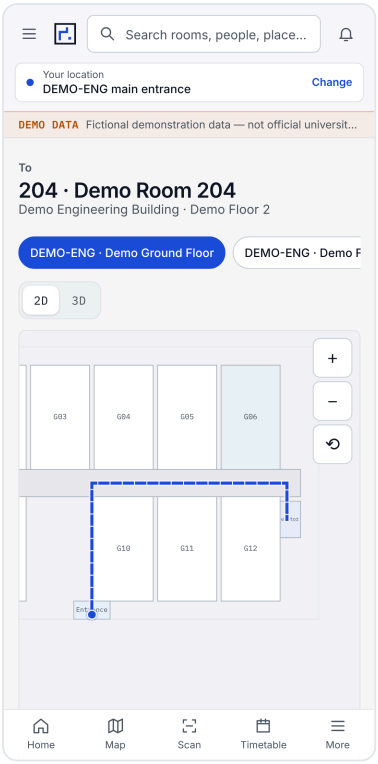
A **pilot in one building** (chosen by the university), then the whole campus once the pilot is approved. Section 5 shows the steps; section 9 lists the decisions we ask the university to make.

2 · WHAT STUDENTS AND STAFF GET

A map-first app that works on every phone



Find a room and see it on its floor.



Directions with route options.

For students and visitors

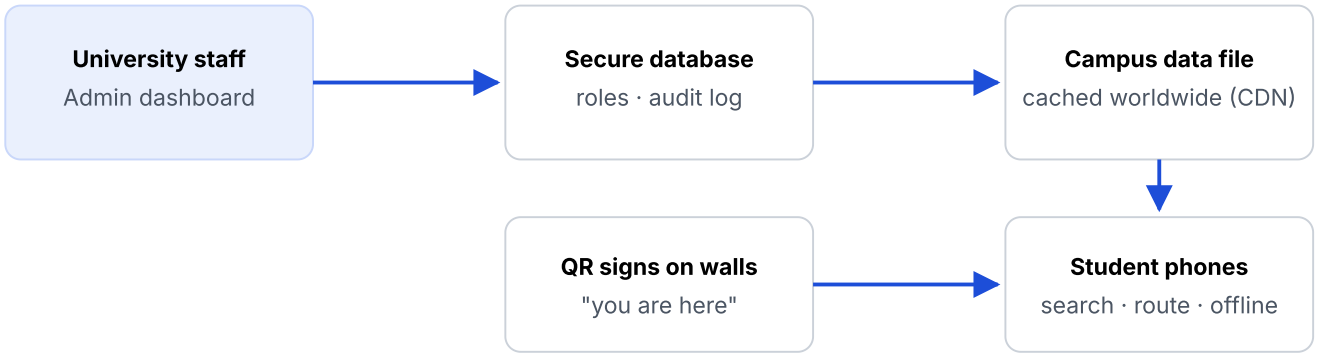
- **Search** rooms, lecturers' offices, departments and buildings — Kurdish (Sorani), Arabic and English, including Kurdish/Arabic digits.
- **Real campus map** (OpenStreetMap) with building outlines, plus **3D building view** with floors and rooms.
- **Directions** through corridors, stairs and elevators. Choose *Shortest*, *Fewer stairs* or *Step-free*.
- **"Where am I?"** by scanning a QR sign on the wall, choosing a place, or tapping the floor plan. No GPS tracking.
- **Nearest** toilets, prayer room, cafeteria, library, by real walking distance.
- **Timetable** by student group, with room changes and cancellations.
- **Works offline** after the first visit — important in basements and busy Wi-Fi times.
- **Report a problem** (wrong room, locked door) in two taps; no name needed.

For the university

- A **staff dashboard** to keep buildings, rooms, maps, timetables and announcements up to date.
- **Temporary closures** (works, exams, broken elevator) that routes avoid automatically.
- **Verification**: only data the team checked on site is shown as "Verified".

3 · HOW IT WORKS

Simple for students, controlled by the university



Maps and directions

- **Outdoors:** the real map (OpenStreetMap). Staff trace each building's outline on it, so position and scale are true to life.
- **Indoors:** each floor plan becomes a network of corridors, doors, stairs and elevators. The app finds the shortest valid path through that network.
- Floor plans are **scaled from a measured distance** (the dashboard has a calibration tool), so distances and walking times are realistic.
- Walking time is always shown as an **estimate**.

Knowing where you are

- GPS does not work reliably inside concrete buildings, so the app uses **QR signs**: scan the sign at an entrance or stairwell and the route starts there.
- Students can also pick a place from a list or tap the floor plan.
- The app **does not track** students and does not need their location permission.

Keeping data correct

- Every change by staff is recorded (who, what, when).
- Students report mistakes; staff fix them and mark them resolved.
- A verification page shows which buildings are complete and checked.

4 · WHAT THE UNIVERSITY NEEDS TO PROVIDE

Information and help we need to install the data

"Essential" items are needed before the pilot. Formats are flexible — paper plans can be scanned; Excel lists are fine.

Item	Why it is needed	Who can usually provide it	Priority
Official approval and a named **coordinator**	To use university names and data, and to reach the right offices quickly.	Presidency	Essential
List of buildings with official codes and names (Kurdish, Arabic, English)	What students search for must match the signs on the buildings.	Planning / Engineering office	Essential
Floor plans for each building (CAD, PDF or paper)	The basis for indoor maps and routes. The biggest piece of work.	Engineering / Projects department	Essential
Room lists: number, name, type, department, capacity	Search results and the timetable link to rooms.	Colleges / Registration	Essential
Confirmation of building locations on the map, and the campus boundary/gates	So buildings appear at their true place. We trace outlines over OpenStreetMap; the university confirms.	Planning office	Essential
Permission to install QR signs at entrances, stairwells and junctions	Lets students start a route from where they stand.	Services / Facilities	Essential
Staff time for a field check (walk each route once)	To confirm that doors, stairs and routes on the map match reality.	Coordinator + 1–2 helpers	Essential
Restricted areas and opening hours (labs, offices, women-only areas, gates)	Routes must never send people through closed areas.	Security / Facilities	Essential
Accessibility information: elevators, ramps, step-free entrances	For correct step-free routes. Ideally tested with students who use wheelchairs.	Facilities / Student affairs	Important
Timetable export (Excel/CSV) and update schedule	"Where is my next lecture?" and room-change notices.	Registration / Colleges	Important
Lecturer public information (name, office, office hours) with consent	Finding a lecturer's office. Only approved, public information.	Colleges / HR	Optional
Official web address and IT contact; hosting/data policy	Trust, and QR signs printed with a permanent address.	IT department	Essential
Data owner per college after launch	Someone must update rooms and closures when things change.	Each college	Important

We will not publish any university information that has not been provided or approved by the university, and anything not yet checked on site is clearly labelled in the app.

5 · HOW THE DATA GETS INSTALLED

From approval to the whole campus

Step 1
Set up
Approval, coordinator, web address, staff accounts.

Step 2
Pilot building
Enter one building fully and place QR signs.

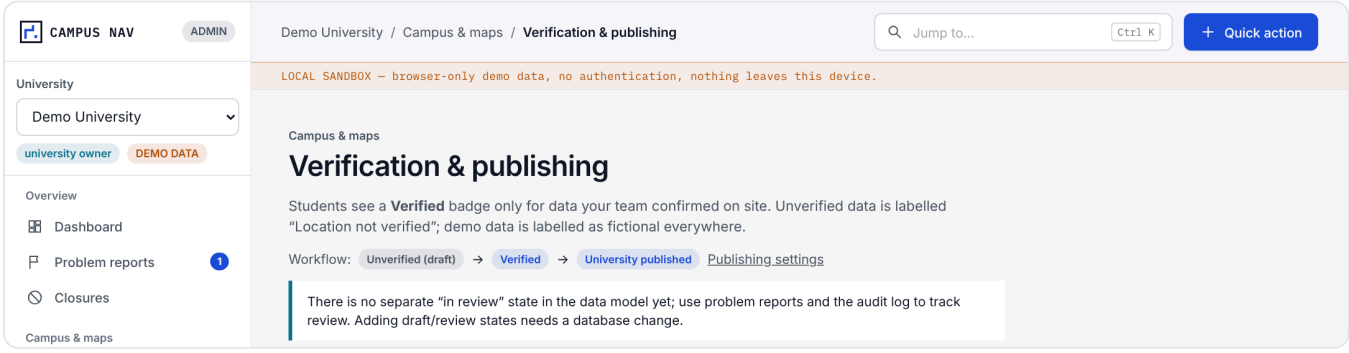
Step 3
Field check
Walk every route; students try it; fix reports.

Step 4
Whole campus
Repeat building by building; launch.

1. **Setup.** The university appoints a coordinator and approves the project. IT provides the web address. We create staff accounts with the right permissions (e.g. college editors can edit only their own rooms).
2. **Place the campus on the real map.** We confirm the campus location and trace each building's outline over the map, then the planning office checks them.
3. **Pilot building.** We draw its floor plans in the dashboard (or import from CAD), calibrate the scale with a tape-measured distance, add rooms, doors, stairs and elevators, and print QR signs.
4. **Field check.** Two people walk every route with the app, write down problems and fix them. Only then is the building marked "Verified".
5. **Student test.** A small group of students uses the app for one or two weeks and sends feedback through "Report a problem".
6. **Campus roll-out.** Repeat for every building, college by college. Timetables are imported each semester.
7. **Maintenance.** Each college keeps its data current; closures and announcements are posted by staff; reports are reviewed weekly.

Time

The time depends mostly on floor-plan availability and the number of buildings. As a planning guide (to be confirmed after the pilot): setup and the pilot building take a few weeks; each further building takes days to a few weeks depending on size and plan quality. We will give a firm schedule once we know the number of buildings and the plan formats.



Staff dashboard — verification shows what is still missing before a building can be marked "Verified". (Demonstration data.)

6 · SERVING ALL STUDENTS

Built so that thousands of students can use it at the same time

The busiest moment is predictable: the start of the semester and the minutes before morning lectures. The design keeps that load away from the database.

How the load is handled

- **One small data file per university.** When a student opens the app, the phone downloads the campus data once. Search and route calculation then run **on the phone itself** — no server work per search or per route.
- **Served from a CDN.** That file and the app itself are cached by a content-delivery network close to the user. The database is asked at most about once every few minutes per university, no matter how many students open the app.
- **Offline after first use.** Returning students load from their phone's storage; the app updates in the background.
- **No student accounts needed** for normal use — nothing to sign in to, nothing to overload.
- Only staff edits and problem reports reach the database, and both are protected and rate-limited.

Measured so far (lab tests)

- Route calculation on a building-sized map: well under a millisecond; a synthetic map with 14,400 points: about 29 ms.
- App start on a slow mobile connection (simulated): about 2 seconds the first time, a fraction of a second afterwards.

Before launch we will

- Run a **load test** that simulates the expected peak (number of students to be agreed with the university) against the real setup.
- Test on ordinary Android phones over campus Wi-Fi.
- Set up uptime and error monitoring with alerts.

These are laboratory measurements on test data, not results from real campus use.

Hosting options

Part	Proposed	Alternative if the university requires local hosting
Student app and campus data file	Global CDN hosting (e.g. Vercel / Cloudflare)	University web server behind a CDN
Database and staff login	Managed PostgreSQL (Supabase) in a nearby region	Self-hosted Supabase / PostgreSQL on university servers
Map background	OpenFreeMap (free OpenStreetMap tiles)	Self-hosted map tiles for the Erbil area

Costs depend on the chosen option and usage and will be quoted separately; the free tiers are usually enough for a pilot.

7 · PRIVACY AND SECURITY

Safe for students, controlled for staff

Students

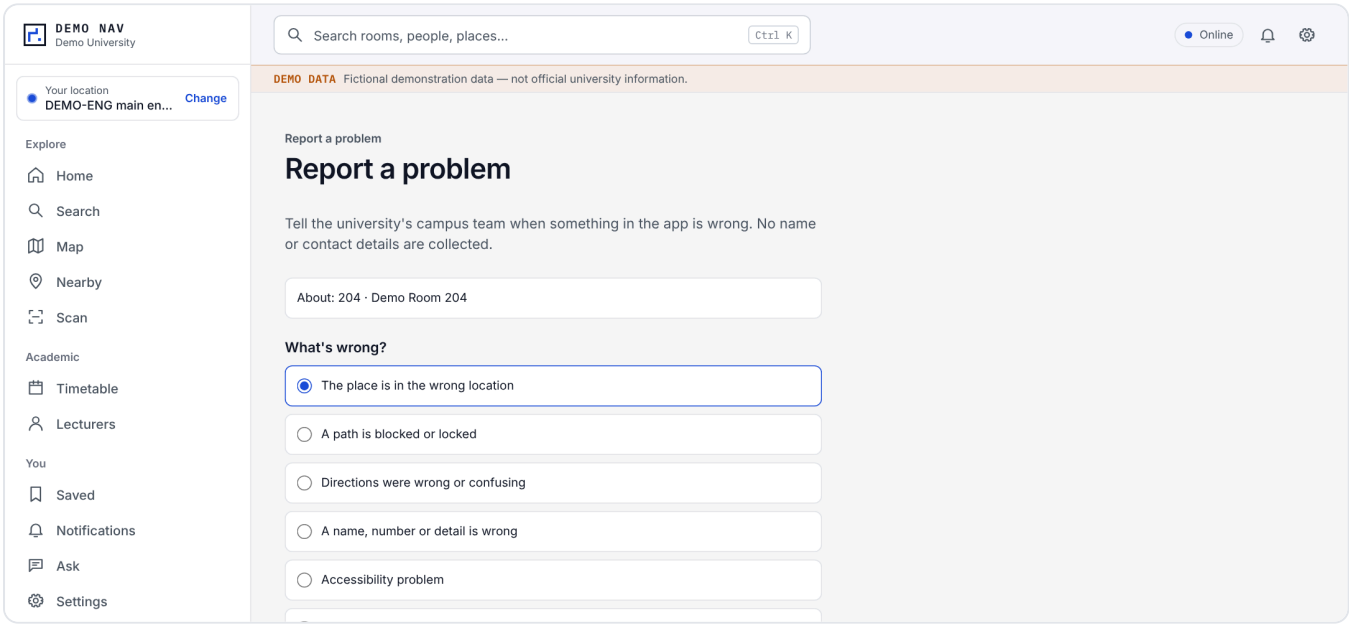
- No account, no tracking, no advertising, no analytics by default.
- No GPS or location permission. Position is only what the student chooses (QR sign or a place on the map), and it stays on their phone.
- Saved places and history stay on the student's own phone; history is off by default.
- Problem reports contain no name or contact details.

Lecturers

- Only approved public information (name, office, office hours) is shown, with consent.

Staff and data

- Staff sign in; roles decide who can change what (e.g. a college editor cannot change another college's rooms).
- The database enforces these rules itself, so they cannot be bypassed from the app.
- Every change is recorded in an audit log.
- Sensitive areas (server rooms, hazardous labs) can be kept off public maps — the university decides what is public.
- Security headers, rate limiting and safe handling of uploaded plans are in place and tested.

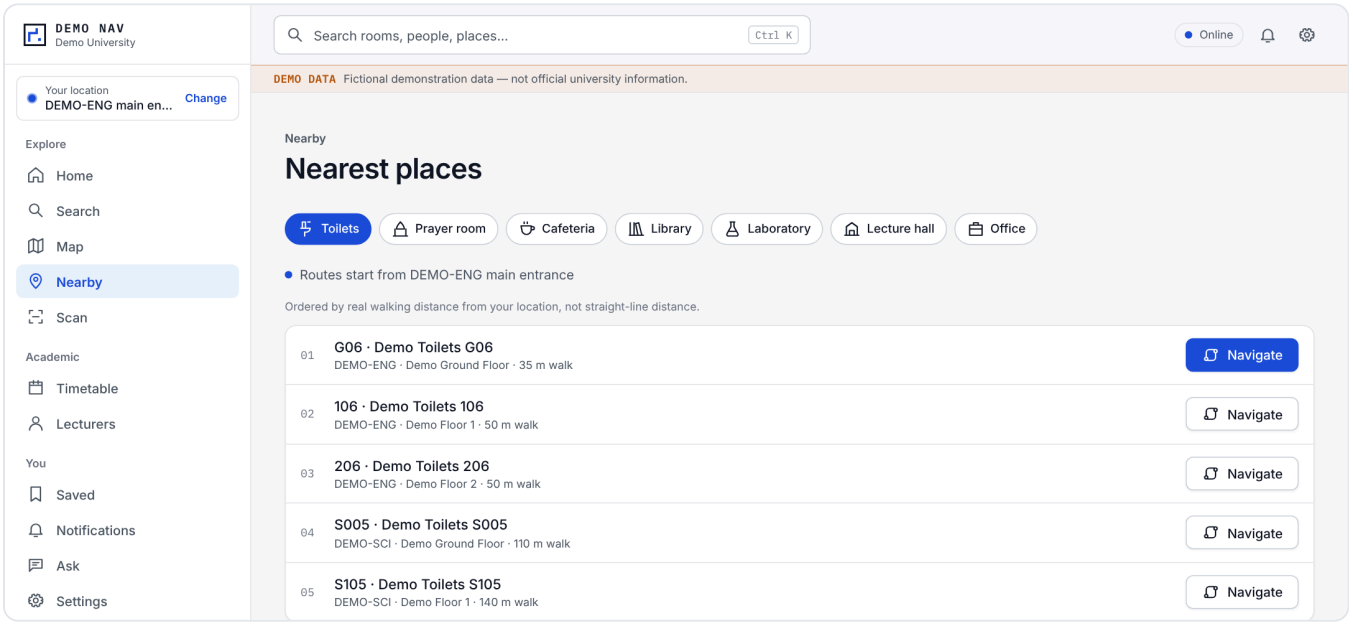


Students can report a wrong room or a locked door without giving any personal information.

8 · HONEST STATUS

What is ready, and what still depends on the pilot

Area	Status	Notes
Student web app (search, maps, directions, timetable, offline, 3 languages)	Built & tested	Automated tests; demonstration data only.
Staff dashboard (data entry, map editor, real-map outlines, QR, closures, reports, verification)	Built & tested	Tested in a local sandbox.
Database security rules	Built & tested	Tested on PostgreSQL; not yet on the production database service.
Salahaddin University data	Not started	Needs the items in section 4.
Accuracy of routes on the real campus	Not yet proven	Will be measured in the field check; we make no accuracy claims before that.
Full-scale load test	Planned	Before launch, with the expected student numbers.
Native phone app (Android / iOS)	Prototype	The web app already works on all phones; store apps can follow.
GPS "blue dot" outdoors, live tracking	Not included	Possible later if the university wants it.



"Nearest" list ordered by real walking distance from the student's position. (Demonstration data.)

9 · DECISIONS REQUESTED

What we ask the university to decide

1. **Approve** a SUL NAV pilot and name a coordinator.
2. **Choose the pilot building** (ideally a busy teaching building with several floors).
3. **Share** the building list, the pilot building's floor plans and room list.
4. **Allow** QR signs in the pilot building and one field-check day.
5. **IT:** provide the official web address and confirm the hosting option (section 6).
6. **Agree** who maintains the data after launch, and which areas must stay off the public map.

What the university gets: a multilingual campus guide for every student and visitor, aimed at reducing time lost looking for rooms and offices; one place for room changes, closures and announcements; and a maintained digital record of the campus layout.

Contact

NULL-TECH — project owner of SUL NAV. Contact details: *to be added by the project owner*.

For Salahaddin University-Erbil — name, position, date

For NULL-TECH — name, date

All screenshots in this document show fictional demonstration data, not Salahaddin University information. Map background: © OpenStreetMap contributors.